

NARESH R

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EDUCATION

M.Sc in Data Science and AI

Birla Institute of Technology and Science, Pilani; **CGPA: 9.1/10**

2025 – Present

Dual Degree - B.Tech in Mechanical Engineering; MBA in Technology

Indian Institute of Technology Madras; **CGPA: 8.2/10**

2020 – 2025

Class XII (CBSE)

Maharishi Vidya Mandir, Chennai; **Percentage: 94.6%**

2019 – 2020

PROFESSIONAL EXPERIENCE

VaultStack AI (Consulting Engagement) - Forward Deployed Engineer

Jun 2026 – Present

Tech stack: Python, Docling, BGE-M3, Qdrant, Cross-Encoder Reranking, Self-hosted LLMs (Ollama), FastAPI, React

- Embedded with client to build an **on-premise, air-gapped** RAG system for financial compliance due diligence, running fully offline on self-hosted LLMs (no external APIs) for strict data confidentiality.
- Building hybrid dense/sparse retrieval (Docling table extraction, BGE-M3, cross-encoder re-ranking) over **scope-partitioned indexes**, preserving table cell structure for figure-level accuracy; designed to scale to **1,000+ documents**.
- Designed a **deterministic scope planner** (resolved in code, model-independent) and a **verifier** enforcing every figure traces to source, with abstention on undisclosed queries.
- Building the full stack (**FastAPI + React**) with page-level citations, plus a staged evaluation harness with hard quality gates across the pipeline.

Accenture - AI Decision Science Analyst

Jan 2025 – Apr 2026

Internal Asset: Intelligent Manufacturing & Production Planning Platform

Tech stack: Python, SQL, Optimization Models, Power BI, Multi-Agent Systems, Azure

- Architected a **multi agent** AI-driven decision platform to enable scenario planning, dynamic analysis, and on-demand insight generation.
- Built a unified application integrating optimization models for capacity planning, resource allocation, and production efficiency.
- Designed persona based dashboards and agent layers, enabling stakeholders to dynamically request analyses, plots, and insights.

Client: Global Fortune 100 Consumer Packaged Goods Manufacturer

Tech stack: Databricks, Azure Data Factory, Python, SQL, Azure

- Led end-to-end health monitoring, debugging, and remediation of enterprise data pipelines supporting global product portfolios.
- Diagnosed and resolved pipeline failures, data quality issues, and performance bottlenecks across Databricks and Azure Data Factory workflows.
- Implemented code-level optimizations and business logic corrections to restore data integrity and ensure SLA adherence for analytics.

Client: World's largest tools manufacturer

Tech stack: Python, SQL, PCA, K-Means, K-Medoids, Power BI

- Owned end-to-end data processing and analytical-pipeline development on large-scale datasets, delivering production-ready customer segmentation.
- Engineered feature selection, PCA dimensionality reduction, and compared clustering approaches (K-Means, K-Medoids) with outlier-capping to improve segmentation stability and interpretability.
- Built Power BI dashboards to present model outcomes to stakeholders; mentored a lateral hire and aligned outcomes with onshore business teams.

Internal Asset: Agentic Long-Range Planning System

Tech stack: Python, CPLEX, Gurobi, HTML, CSS, JavaScript, React, Django, APScheduler, WebSockets

- Built **MILP-based** supply chain planning tools using Python, CPLEX, and Gurobi with a Django backend and React frontend.
- Implemented **async job orchestration** with APScheduler and WebSockets for real-time scenario execution and live status updates.
- Integrated AI agents and interactive Recharts dashboards for no-code scenario execution and decision support.

Razorpay - Product Development Intern

Jul 2024 – Oct 2024

Tech stack: Google Apps Script, Python, Liquid, CSS, HTML, JavaScript

- Automated personalized PDF report generation for D2C Summit participants; each report delivered in **under 1 minute**.
- Integrated Razorpay scripts for **200+ merchants**, resolved script conflicts, and implemented A/B testing to improve performance.
- Developed an admin dashboard feature reducing manual work by **4 hours per merchant**, boosting operational efficiency.

TVS Electronics - Machine Learning Intern

May 2024 – Jul 2024

Tech stack: Python, AWS Bedrock, MySQL, Llama 3, Streamlit

- Cleaned and enhanced a large volume of service data using basic preprocessing, NLP, and Llama 3, improving data quality and providing a reusable notebook for future text cleaning.
- Utilized **Retrieval-Augmented Generation (RAG)** to retrieve and generate relevant solutions from service data efficiently.
- Developed a prototype Streamlit web app that efficiently retrieves and displays top solutions, streamlining operational workflows.

Tecnod8 - Generative AI Intern

Apr 2024 – May 2024

Tech stack: Python, Pandas, BeautifulSoup, PyPDF2, YouTube API, OpenAI API, Vertex AI

- Designed and implemented a layered approach to enhance the efficiency and comprehensiveness of data collection.
- Developed scripts to automate the extraction of data from PDF documents, increasing the speed and accuracy of data preprocessing and extracted relevant data from YouTube videos, converting audiovisual content into structured data.
- Transformed extracted data into a format suitable for instruction fine-tuning, resulting in the successful **fine-tuning of the Llama**.

Razorpay - Software Development Intern

Jun 2023 – Aug 2023

Tech stack: MS Excel, Excel VBA, Google Suite, Google Apps Script, JavaScript

- Achieved a **95%+ reduction** in data errors for a 1000+ entry Excel file using innovative VBA scripting, cutting weekly error rates from **20-25% to under 5%**, significantly elevating data accuracy and computational efficiency.
- Saved **2-3 hours per on-call report** for engineers by automating report generation, ensuring heightened precision. Expanded this automation across diverse teams using various reporting tools as a stretch goal.
- Reduced average resolution time by **30%** and optimized on-call bandwidth by developing Call-to-Action feature within the dashboard.

Royal Enfield - Summer Intern

May 2022 – Jul 2022

Tech stack: MS Office, AutoCAD

- Achieved cost savings of **10 Lakhs per annum** by proposing process streamlining and space optimization ideas through data analysis and insights gained from hands-on experience with various business systems and data extraction

PUBLICATIONS & CONFERENCES

- *GenAI-Driven Intelligent Email Ticketing for Sentiment-Aware Customer Support*. Proceedings of the Sixth International Conference on Advances in Computer Engineering and Communication Systems (ICACECS 2025), Springer Nature, Smart Innovation, Systems and Technologies. Authors: Sreeja S; **Naresh R**; L. C. S. Reddy.
- *Optimizing Delivery Operations in Dynamic Urban Environments: A Comprehensive Model*. Presented at the XXVII Annual International Conference of the Society of Operations Management (SOM) 2024. Authors: Sreeja Sunkeswaram; **Naresh R**; G. Viswanath; C. Rajendran.
- *Enhancing Sentiment Analysis on Amazon Reviews Using Ensemble Learning Models*. International Journal of Innovative Research in Technology, 11(8), 3349–3354, 2025. Authors: Sunkeswaram Sreeja; **Naresh R**.
- *Automated Assessment of Fruit Quality Using Transfer Learning and MobileNetV2*. International Journal for Research in Applied Science & Engineering Technology (IJRASET), 13(IV), 166–172, 2025. Authors: Bira Odra; Sunkeswaram Sreeja; **Naresh R**.

PROJECTS & COMPETITIONS

- **Automated Grading and Feedback System**
Achieved **second runner-up among 104+ teams**. Developed an automated grading system tailored to descriptive questions. Leveraged a **Panel of LLM Evaluators (PoLL)** for rigorous, unbiased assessment. Implemented a robust grading rubric, enhancing transparency and mitigating scoring inconsistencies. Delivered detailed, personalized feedback to students.
- **Real-Time Streaming Pipeline for Food Orders**
Engineered a scalable streaming system using Kafka, Spark, and PostgreSQL to process **300+ incremental CDC events** in real time. Automated ingestion → transformation → data lake storage using Parquet, achieving reliable micro-batch processing and verifiable downstream analytics.
- **Community Detection and Location Routing**
Guide: Prof. Viswanath, Adjunct Faculty, DoMS, Indian Institute of Technology Madras & Principal Research Scientist, TCS. Engineered dynamic clustering algorithms to pinpoint optimal depot locations using real-time synthetic population data. Deployed advanced routing strategies with synthetic data, reducing delivery times and the number of delivery partners.
- **AMEX Hackathon**
Performed data processing and feature extraction to predict cricket match winners. Developed a hybrid model using LightGBM, XGBoost, and CatBoost, achieving **82% accuracy** on test data.
- **Surface Defect Detection**
Built high-accuracy defect detection systems using OpenCV, Keras, and TensorFlow. Achieved **98% accuracy**.

CERTIFICATIONS

- AWS Certified AI Practitioner
- AWS Certified Cloud Practitioner
- Reinvention with Agentic AI - Accenture
- Amazon ML Summer School
- IBM Quantum Computing Challenge 2024
- Gen AI Pinnacle Program - Analytics Vidhya

SKILLS

Languages / Tools: Python, C++, C, SQL, R, JavaScript, TypeScript, Bash, Git, Gurobi, CPLEX, Power BI, Tableau

GenAI / LLM: RAG, Vector Databases (Qdrant, FAISS), Embeddings, Re-ranking, Self-hosted LLMs (Ollama), LangChain, Hugging Face

ML / Backend: PyTorch, TensorFlow, Keras, Scikit-learn, FastAPI, Flask, Django, React, REST APIs, WebSockets, Multi-Agent Systems

Deployment / MLOps: On-Prem & Air-Gapped Deployment, Docker, Kubernetes, CI/CD (GitHub Actions), Grafana, Linux, GPU/CUDA, AWS, Azure, Databricks, Apache Spark, Apache Kafka